What's new

Rotating needles in space: the road to the Kakeya conjecture, and why it matters

Terence Tao · 25 August 2026

https://terrytao.wordpress.com/2026/08/25/rotating-needles-in-space-the-road-to-the-kakeya-conjecture-and-why-it-matters/

tao2typ · converted to Typst 2026-10-14 (UTC)

A recent tradition of the ICM is to have a non-technical popular article written for each of its medallists. (This is separate from the older tradition of having a laudatio for the winner, which is similar but aimed at a more advanced audience.) I was approached to write such an article for Hong Wang on the occasion of her Fields Medal. I have uploaded my initial version of this article, "Rotating needles in space: the road to the Kakeya conjecture, and why it matters", to the arXiv. Chris Sogge gave the corresponding laudatio; both should eventually appear in the Proceedings of the ICM.

Comments (6)

Anonymous · 26 August 2026

Thanks for sharing this. It was a great read for a non-mathematician and I now have a better understanding of Wang and colleagues' contribution and impact on so many seemingly unrelated fields!

Anonymous · 26 August 2026

Thank you for this accessible and explanatory article! I am very interested in the Gibbs phenomenon mentioned in the article, and I am wondering why this phenomenon occurs. May I ask what the fundamental cause of the Gibbs phenomenon is?

Anonymous · 26 August 2026

The word "rangle" appears on page 3, was that a typo?

[That should be "rectangle"; it will be corrected in the next revision of the ms -T.]

Anonymous · 31 August 2026

page 11, "to difficult", typo?

[Thanks, this will be corrected in the next revision of the ms. -T]

Justin Bian · 1 September 2026

Dear Professor Tao,

In the "back of the envelope calculation", at the end of page 3, we have "a typical point in one rectangle is covered by only one or two other rectangles". Can we pin down the number to one or two, or should we also use O(1) wordings here?

[I will reword "one or two" to "a small number of" in the next revision – T.]

Anonymous · 4 September 2026

I am curious about the use of Al assistance to generate images in published articles (though not necessarily this article in particular). Some images, like figure 1 in your article, can easily be created using Al-generated TikZ code. But more complicated images (perhaps figure 5—the Gaussian wave packet) may be generated as an image and then imported into the file.

How does an author prevent unintentional plagiarism and/or copyright infringement in such cases? It seems possible that an Al-created image could be "similar enough" to some other published image to warrant this concern.